

MP SuperServer SYS-681E-TR

6U 8-socket system with 128 DIMMs, 24 NVMe/SAS/SATA drive bays, and 24 PCIe slots



[More details here](#)

Key Applications

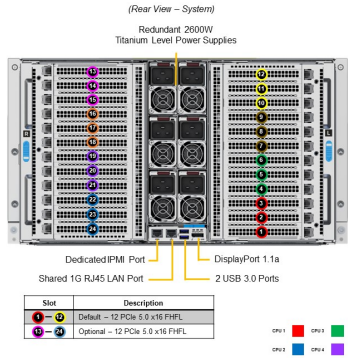
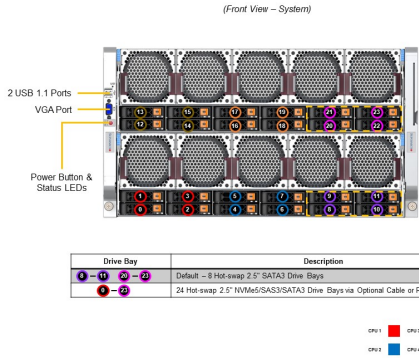
In-Memory Database, Virtualization, ERP, CRM, Research Lab/National Lab,
Scale-up HPC,

Key Features

- Octa Socket E (LGA-4677) 4th Generation Intel® Xeon® Scalable processors;
- 128 DIMM Slots; Up to 32TB DDR5-4800MHz ECC RDIMM/RDIMM-3DS;
- Optional PCIe configurations up to 24 PCIe slots with support for 12 double-width GPU/Accelerator cards;
- 24x 2.5" Hot-swap NVMe/SAS3/SATA3 drive bays; 2x internal M.2 NVMe/SATA; Optional RAID support via storage add-on card;
- 10 heavy duty fans w/ optimal fan speed control;



Form Factor	6U Rackmount Enclosure: 449 x 265 x 841mm (17.68" x 10.4" x 33.1") Package: 720 x 922 x 1080mm (28.34" x 36.3" x 42.5")
Processor	8 Socket E (LGA-4677) 4th Gen Intel® Xeon® Scalable processors Up to 60C/120T; Up to 112.5MB Cache per CPU
GPU	Up to 12 double-width GPUs
System Memory	Slot Count: 128 DIMM slots Max Memory (1DPC): Up to 16TB 4800MT/s ECC DDR5 RDIMM Max Memory (2DPC): Up to 32TB 4400MT/s ECC DDR5 RDIMM
Drive Bays Configuration	Default: Total 8 bays • 8 front hot-swap 2.5" SATA drive bays Option A: Total 24 bays • 24 front hot-swap 2.5" NVMe*/SAS*/SATA* drive bays (*NVMe/SAS/SATA support may require additional storage controller and/or cables, please see the optional parts list for details) M.2: 2 M.2 PCIe 3.0 x4 NVMe/SATA slots (M-key 2280/22110)
Expansion Slots	PCI-Express (PCIe) Configuration: Default • 12 PCIe 5.0 x16 FHFL slots • 2 PCIe 5.0 x16 LP slots Option A* • 12 PCIe 5.0 x16 FHFL slots M.2: 2 M.2 NVMe/SATA slots (M-key 2280/22110)
On-Board Devices	1 RJ45 1GBASE-T
Input / Output	LAN: 1 RJ45 1 GbE Dedicated BMC LAN port 1 RJ45 1 GbE LAN port USB: 2 USB 3.0 ports(rear) 2 ports(front) Video: 1 VGA port(Front) 1 DisplayPort port(Rear)



System Cooling	Fans: Up to 10 Heavy duty counter-rotating 8cm Fan(s) Air Shroud: 1 Air Shroud
Power Supply	4x 2600W Redundant (2 + 2) Titanium Level (96%) power supplies
System BIOS	BIOS Type: AMI 32MB SPI Flash EEPROM
Management	SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!
PC Health Monitoring	CPU: 4+1 Phase-switching voltage regulator Monitors for CPU Cores, Chipset Voltages, Memory FAN: Fans with tachometer monitoring Pulse Width Modulated (PWM) fan connectors Status monitor for speed control Temperature: Monitoring for CPU and chassis environment Thermal Control for fan connectors
Dimensions and Weight	Weight: Gross Weight: 242.5 lbs (110 kg) Net Weight: 176.37 lbs (80 kg) Available Color: Black
Operating Environment	RoHS Compliant Operating Temperature: 10°C to 35°C (50°F to 95°F) Non-operating Temperature: -40°C to 60°C (-40°F to 140°F) Operating Relative Humidity: 8% to 90% (Non-Condensing) Non-operating Relative Humidity: 5% to 95% (Non-Condensing)
Motherboard	Super X13OEI-CPU
Chassis	CSE-618TS-R5K26P